Coding Assignment: DynaQ and DynaQ + in a Changing Maze

Course Information

- Course: Reinforcement Learning
- Topic: Dyna-Q and DynaQ +
- Language: Python 3
- Difficulty: Intermediate
- Estimated Time: 2–3 hours

Learning Objectives

After completing this assignment, students should be able to:

1. **Implement the Dyna-Q algorithm** by integrating direct reinforcement learning, model learning, and planning into a single agent.
2. **Implement the Dyna-Q+ algorithm** by incorporating an exploration bonus into the planning process to encourage continued exploration.
3. **Differentiate between real experience and simulated experience**, and explain how Dyna-Q uses both to improve learning efficiency.
4. **Explain the role of planning** in model-based reinforcement learning and how the number of planning steps affects learning speed and computational cost.
5. **Analyze the effect of environmental changes** on reinforcement learning agents and explain why model-based agents must continue to explore in non-stationary environments.
6. **Compare the performance of Dyna-Q and Dyna-Q+** using cumulative reward curves and interpret the differences in their learning behavior.
7. **Evaluate the exploration–exploitation tradeoff** in Dyna-Q and Dyna-Q+, particularly the impact of the exploration bonus in Dyna-Q+.
8. **Design and conduct reinforcement learning experiments**, including selecting hyperparameters, running controlled comparisons, collecting performance metrics, and visualizing results.
9. **Interpret experimental results** to explain why Dyna-Q+ adapts more quickly than Dyna-Q after changes in the environment.

10. **Develop proficiency in implementing model-based reinforcement learning algorithms in Python**, including representing environments, maintaining a learned model, and performing planning updates.

Coding Question: Dyna-Q vs. Dyna-Q+ in a Changing Maze

A 6×9gridworld initially contains a path from the start state S to the goal state G. After 500 time steps, the maze changes by opening a shortcut to the goal.

Implement Dyna-Q and Dyna-Q+ agents and compare their performance.

Maze Layout and Parameters

Maze Specification

The maze is a 6-row × 9-column grid (0-indexed, with row 0 at the top).

- Start state (S): row 5, column 3
- Goal state (G): row 0, column 8
- Reward: +1 for the transition into G; 0 for every other transition
- Wall before step 500: blocks columns 1–8 of row 3, leaving only column 0 open — (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (3,7), (3,8)
- Wall after step 500: shortens by one cell on the right, leaving column 0 open (as before) and opening a new shortcut at column 8 — (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (3,7)

Maze layout before step 500 (wall spans columns 1–8 of row 3):

	0	1	2	3	4	5	6	7	8
0									**G**
1									
2									
3									
4									
5				**S**					

Maze layout after step 500 (wall spans columns 1–7 of row 3; column 8 opens as the new shortcut, column 0 remains open):

	0	1	2	3	4	5	6	7	8
0									G
1									
2									
3									
4									
5				S					

Legend: gray = wall, green (S) = start, gold (G) = goal, white = open cell. Rows and columns are 0-indexed; row 0 is the top of the maze.

Hyperparameters

- Discount factor: $\gamma = 0.95$
- Learning rate: $\alpha = 1.0$
- Exploration rate: $\varepsilon = 0.1$
- Planning steps: $n = 10$ (already specified above under Tasks)
- Exploration bonus constant: $\kappa = 0.001$ (already specified above under Tasks)

Tasks

- Implement the Dyna-Q algorithm with 10 planning steps.
- Implement the Dyna-Q+ algorithm with 10 planning steps and an exploration bonus

$$\kappa \sqrt{\tau},$$

where:

  - → $\kappa = 0.001$,
  - → τ is the number of time steps since the state-action pair was last executed.
- Run both agents for 3000 time steps.
- Change the maze at time step 500 by opening a shortcut.
- Plot the cumulative reward versus time for both algorithms.
- Briefly explain why one algorithm adapts faster after the shortcut appears.

Model initialisation for Dyna-Q+

The exploration bonus $\kappa\sqrt{\tau}$ is applied only during planning, and planning can only sample state-action pairs that exist in the model. If the model only contained transitions the agent had actually executed, an action that was never tried would never be planned over, would never receive a bonus, and the agent would have no incentive to explore it. Following Sutton and Barto (2018, Section 8.3, footnote on Example 8.3), the first time the agent visits a state s, all four actions from s are added to the model as if they had been tried once and returned to s with reward 0. The time-since-last-taken counter τ for these entries starts at the beginning of the run, so their bonus grows until the action is actually executed, at which point the model entry is overwritten with the real transition. This initial "experience" is what allows the bonus to pull the agent toward the shortcut after the maze changes. Without it, Dyna-Q+ performs worse than Dyna-Q. Dyna-Q does not use this initialisation.

Conceptual Question

Why does Dyna-Q+ outperform Dyna-Q after the maze changes?

Grading Rubric

Category	Points	Evaluation Criteria
Maze Environment	15	Correct implementation of the changing maze, including start state, goal state, walls, valid actions, and opening the shortcut after the specified time step.
Dyna-Q Implementation	25	Correct implementation of Dyna-Q, including Direct-RL (Q-learning) update, learned model, planning loop, and ε-greedy action selection.
Dyna-Q+ Implementation	20	Correct implementation of Dyna-Q+, including exploration bonus (r+ \kappa\sqrt{\tau}), tracking (\tau), and applying the bonus only during

Category	Points	Evaluation Criteria
		planning.
Experimental Evaluation	15	Runs both algorithms under identical conditions, collects cumulative rewards, and changes the maze at the specified time step.
Results Visualization	10	Produces a clear plot of cumulative reward vs. time with appropriate labels, legend, and indication of when the maze changes.
Analysis & Discussion	15	Correct explanation of why Dyna-Q+ adapts faster after the shortcut appears and discussion of exploration vs. exploitation.